ModelIt!

Teacher's Guide

The Ocean Is Turning to Acid

How CO2 Emissions Are Dissolving Marine Life

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-ESS3-6, HS-PS1-7
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-6, HS-PS1-7

Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity. Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction.

Three-Dimensional Learning

SEP	**Using Mathematics and Computational Thinking** — Students use computational models to represent the quantitative relationships between CO_2 emissions, ocean absorption, pH changes, and biological impacts.
DCI	**ESS3.C: Human Impacts on Earth Systems / PS1.B: Chemical Reactions** — Human CO_2 emissions are absorbed by the ocean, driving chemical reactions that produce carbonic acid and lower pH. Atoms are conserved — the carbon in your exhaust becomes the acid dissolving coral.
CCC	**Stability and Change** — The ocean's pH has been stable for millions of years. Human CO_2 emissions have disrupted this stability faster than marine organisms can adapt, pushing the system toward a new — and potentially catastrophic — equilibrium.

CAST Testing Alignment

TRACE:

Follow a carbon atom from gasoline in a car engine, through the atmosphere, into the ocean, and into a chemical reaction with seawater. Write the chemical equations and explain each step.

EVALUATE:

A policy advisor claims that 'the ocean is cleaning up our CO_2 for free.' Using your model, explain why this statement is dangerously misleading.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Ocean acidification is often called 'the other CO2 problem' — less famous than global warming but potentially just as devastating. The chemistry is straightforward, the evidence is irrefutable, and the consequences for marine life are already visible.

The Chemistry of Ocean Acidification

When CO2 dissolves in seawater, it reacts with water to form carbonic acid: CO2 + H2O → H2CO3. This carbonic acid then dissociates, releasing hydrogen ions (H+) that lower pH: H2CO3 → H+ + HCO3-. The excess hydrogen ions also react with carbonate ions ($CO3^{2}$-) that marine organisms need for building shells: H+ + $CO3^{2}$- → HCO3-. This is why acidification is a double hit — it increases acidity AND removes the building blocks organisms need.

The Scale of the Problem

The ocean has absorbed approximately 525 billion tons of CO2 from the atmosphere since the Industrial Revolution — about 30% of all human emissions. Surface ocean pH has dropped from 8.2 to 8.1. While 0.1 sounds tiny, pH is logarithmic: this represents a 30% increase in hydrogen ion concentration. Current projections show pH could drop to 7.8 by 2100 — a 150% increase in acidity from pre-industrial levels.

Biological Impacts

Organisms that build calcium carbonate structures — corals, oysters, clams, sea urchins, and many plankton species — are most vulnerable. Below a pH threshold, their shells and skeletons literally dissolve faster than they can build them. Coral reefs, which support 25% of all marine species despite covering less than 1% of the ocean floor, are facing dissolution. Pteropods (sea butterflies) at the base of many food chains are already showing shell thinning.

The Recovery Problem

Even if CO2 emissions stopped completely today, ocean acidification would continue for decades as existing atmospheric CO2 continues dissolving. Full recovery of ocean pH would take tens of thousands of years because the deep ocean circulation that mixes surface and deep water operates on millennial timescales. This makes prevention far more effective than cure.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify CO2 emissions, ocean CO2 absorption, ocean pH, and coral health as components of the ocean acidification system.

E **Establish Relationships**
Students discover that emissions drive absorption (+), absorption drives pH down (-), and lower pH destroys coral health (-).

V **Visualize & Model**
Students build a model showing the causal chain from human emissions to marine ecosystem destruction.

E **Evaluate Outcomes**
Students run current trajectory, Paris Agreement, and zero emission scenarios to compare outcomes.

R **Revise & Extend**
Students add water temperature, biological buffering, or deep ocean circulation to explore more complex dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This coral reef is dissolving. Not from pollution you can see — from an invisible gas. How?"*

Watch For

Show before/after images of coral bleaching. Ask: What could cause solid rock to dissolve?

Slide 2: Learning Objectives

3 min

Say: *"Today we're tracing an invisible path: from your car's tailpipe to a coral reef 10,000 miles away."*

Watch For

Have students read objectives. Pre-teach pH with a visual scale.

Slide 3: Big Question

3 min

Say: *"How does burning gasoline in your city dissolve a shell on the ocean floor?"*

Watch For

Show the chemical pathway. Ask: Where does exhaust CO2 go?

Slide 4: LEVER Framework

2 min

Say: *"We'll model how CO2 travels from atmosphere to ocean and what it does when it arrives."*

Watch For

Preview LEVER steps. Emphasize we're tracking a MOLECULE's journey.

Slide 5: Activity 1: Components

8 min

Say: *"What can humans control? What's a consequence of our choices?"*

Watch For

Guide sorting. CO_2 Emissions is the only truly external component — everything else is a downstream effect.

Slide 6: Activity 2: Connections

8 min

Say: *"More CO_2 absorbed means lower pH. But what does lower pH mean for living things?"*

Watch For

Show the chemical equation. Demonstrate with vinegar on chalk (calcium carbonate).

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test three futures: do nothing, follow Paris, or go to zero. Can the ocean recover?"*

Watch For

Students predict pH for each scenario. The zero-emission recovery delay is the key insight.

Slide 8: Discoveries

5 min

Say: *"Even if we stopped ALL emissions today, the ocean keeps acidifying for decades. Why?"*

Watch For

Discuss the lag effect. Existing atmospheric CO_2 keeps dissolving.

Slide 9: STEM Challenge

12 min

Say: *"A Caribbean reef is dying. Design a rescue plan that addresses both symptoms AND causes."*

Watch For

Groups design multi-pronged strategies: immediate protection + long-term emission reduction.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** CO_2 Emissions
- **INTERNAL:** Ocean CO_2 Absorption, Ocean pH, Coral Health

Relationships

- **CO_2 Emissions to Ocean CO_2 Absorption:** POSITIVE (+) — More atmospheric CO_2 means more dissolves into the ocean. The ocean absorbs roughly 30% of our emissions — more emissions means more absorption.
- **Ocean CO_2 Absorption to Ocean pH:** NEGATIVE (-) — More dissolved CO_2 creates more carbonic acid, releasing hydrogen ions that lower pH. Every additional ton of CO_2 absorbed makes the ocean measurably more acidic.
- **Ocean pH to Coral Health:** POSITIVE (+) — Higher pH (less acidic) supports healthy coral growth. As pH drops, coral's calcium carbonate structures dissolve faster than organisms can build them, leading to reef death.

Key Discoveries

- The ocean acts as a massive CO_2 sink — absorbing billions of tons yearly — but this absorption comes at the cost of increasing acidity
- Ocean pH has dropped 0.1 units since pre-industrial times — a 30% increase in hydrogen ion concentration
- Even under the Paris Agreement, ocean acidification continues for decades because CO_2 already in the atmosphere keeps dissolving
- Coral reefs and shellfish are the first casualties because their calcium carbonate structures literally dissolve in more acidic water

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The ocean is basic so it can handle some acid"	While seawater is indeed slightly basic (pH 8.1), marine life evolved over millions of years in this exact pH range. A shift of 0.1 pH units represents a 30% change in acidity — equivalent to changing the oxygen concentration in air by 30%. Organisms can't just 'handle' that.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"CO2 is natural so it can't be harmful"	CO2 is natural and essential for photosynthesis. The problem isn't CO2 existing — it's the RATE of increase. Natural CO2 cycles take thousands of years; we've increased atmospheric CO2 by 50% in 150 years. The ocean can absorb CO2 gradually, but not at this pace. It's like the difference between rain (natural) and a flood (too much, too fast).
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We can just add chemicals to fix the ocean"	The ocean contains 1.335 billion cubic kilometers of water. To raise its pH by 0.1 units would require billions of tons of alkaline minerals. We'd need to mine, transport, and distribute more material than the entire global mining industry produces. And we'd need to keep doing it every year. Prevention (cutting emissions) is millions of times more practical than cure.